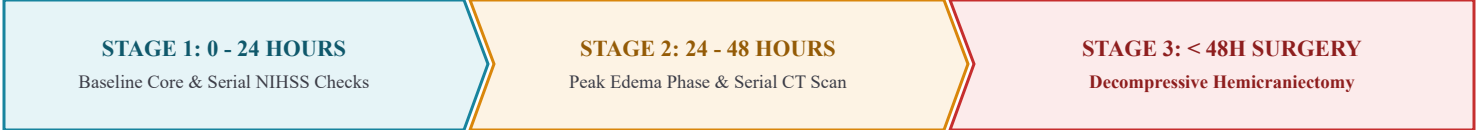


# Malignant Infarction

Decompressive hemicraniectomy selection criteria, evidence, and supportive ICU care.



## 1. Decompressive Hemicraniectomy Selection Criteria

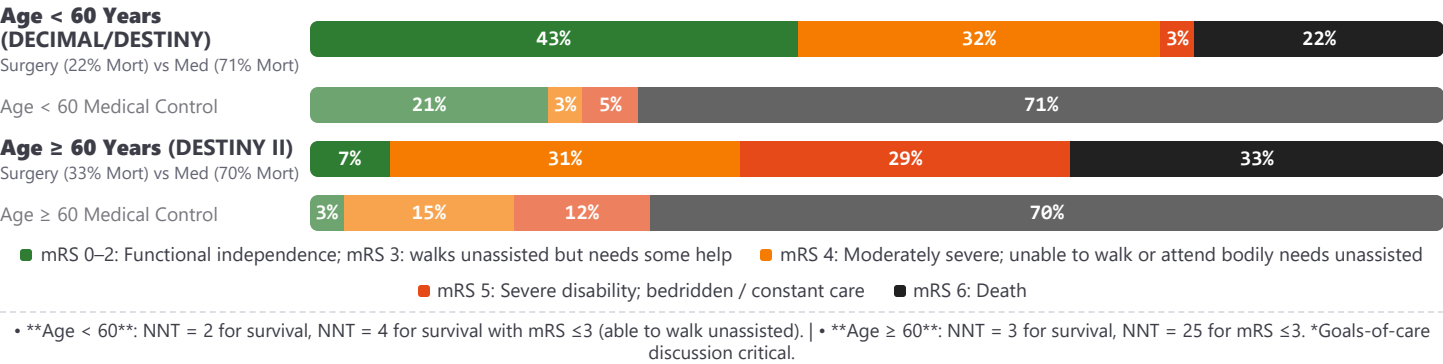
**Clinical Deficit Severity:**

- NIHSS > **15** (non-dominant hemisphere)
- NIHSS > **20** (dominant hemisphere)
- AND decrease in level of consciousness (NIHSS Item 1a score  $\geq 1$  / obtunded or stuporous)
- \*\*Timing\*\*:** Surgery performed **within 48 hours** of onset.

**Radiographic Markers:**

- Infarction of  $\geq 50\%$  of the MCA territory (CT/MRI)
- DWI core volume > **82 mL** within 6 hours
- DWI core volume > **145 mL** within 14 hours
- Midline shift or mass effect on repeat imaging
- \*\*Surgical Spec\*\*:** Bone flap diameter  $\geq 12\text{--}15\text{ cm}$  with duraplasty.

## 2. Surgical Outcomes & Evidence (By Age Group)



## 3. Supportive ICU Care & Medical Management

|             |                                                                                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positioning | Elevate HOB 30 degrees; maintain straight head/neck alignment to maximize venous outflow.                                                                                                   |
| Fluids      | Maintain euvolemia with isotonic fluids. <b>Avoid hypotonic fluids</b> (e.g. D5W, 0.45% NS) that can worsen edema; balanced crystalloids such as LR should follow local neuro-ICU protocol. |
| Osmotherapy | Consider <b>targeted PRN</b> hyperosmolar agents (HTS 3% or Mannitol) for acute decline or severe mass effect. <i>Prophylactic osmotherapy is not recommended.</i>                          |
| Metabolic   | Target normothermia ( $<37.8^{\circ}\text{C}$ ). Target normocapnia (PaCO <sub>2</sub> 35-45 mmHg); avoid hypoventilation/hypercapnia.                                                      |
| Steroids    | <b>Class III (Harmful):</b> Corticosteroids are NOT recommended for reducing cerebral edema in acute ischemic stroke.                                                                       |

**DECIMAL:** Vahedi K et al. *Stroke*. 2007;38:2506-2517. | **DESTINY:** Jüttler E et al. *Stroke*. 2007;38:2518-2525.  
**HAMLET:** Hofmeijer J et al. *Lancet Neurol*. 2009;8:326-333. | **DESTINY II:** Jüttler E et al. *N Engl J Med*. 2014;370:1091-1100.  
**AHA Guidelines:** Wijdicks EF et al. *Stroke*. 2014;45:1222-1238.